HOSTEL MAINTENANCE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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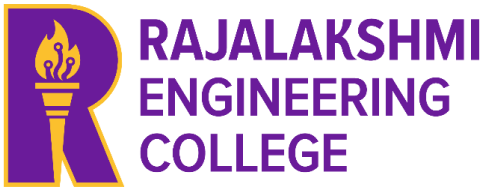
in partial fulfilment of the End Semester Examination for the course

CB23F35 – LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled HOSTEL MAINTENANCE MANAGEMENT SYSTEM is the bonafide work of AKSHAYA S (2116241401009), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, under the guidance of the course faculty.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, AKSHAYA S (2116241401009), hereby declare that the project report titled HOSTEL MAINTENANCE MANAGEMENT SYSTEM, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of the project work carried out by me. The application described in this report is based on the forms, reports, navigation sections and dashboard demonstrated in the supplied Zoho Creator project screenshots.

I further declare that the report has been prepared for academic submission and that the project features described as implemented are limited to those evidenced by the application screens and records supplied for this report.

Place: Chennai

SIGNATURE OF THE STUDENT

Date: 2/11/2026

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AKSHAYA S

ABSTRACT

Hostel maintenance involves maintaining student records, staff information, room allocations and maintenance complaints. When these activities are recorded in separate registers or spreadsheets, administrators may find it difficult to retrieve information, monitor pending work and review complaint resolution. A centralised digital application can improve the organisation and visibility of these activities.

The Hostel Maintenance Management System was developed using Zoho Creator to organise student records, staff details, hostel room information, maintenance complaints and feedback in one application. The available list views support record review, while the dashboard provides a consolidated view of complaint status and category distribution. The supplied dashboard screenshot shows 20 complaints, 15 students and 10 staff. The project demonstrates how a low-code platform can support hostel maintenance administration.

The supplied application screenshots demonstrate data-entry forms, list/report views and dashboard components. The dashboard displays 20 total complaints, 6 pending complaints, 5 complaints in progress and 9 resolved complaints, together with 15 students and 10 staff. The category chart displays Electrical 6, Plumbing 5, Furniture 3, Cleaning 3, Internet 2 and Other 1.

Keywords: Hostel Maintenance; Zoho Creator; Low-Code Development; Student Records; Staff Management; Hostel Rooms; Maintenance Complaints; Complaint Feedback; Dashboard.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Hostel Maintenance Management System
Application Domain	Hostel Administration and Maintenance Management
Student Name	Akshaya S
Register Number	2116241401009
Course	CB23F35 – Low-Code Application Development using Zoho Creator
Platform	Zoho Creator
Environment	Development
Main Modules	Student Details; Staff Details; Hostel Rooms; Maintenance Complaints; Complaint Feedback; Hostel Admin Dashboard
Student Records	5 visible in the supplied list; dashboard shows 15
Staff Records	5 visible in the supplied list; dashboard shows 10
Hostel Room Records	11
Total Complaints	20
Pending Complaints	6
In Progress	5
Resolved Complaints	9
Feedback Records	1 visible record

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Hostel maintenance is an important part of residential campus administration. A hostel office must maintain student details, room allocations, staff information and records of facility-related complaints. Common issues include electrical faults, plumbing leaks, cleaning requirements, internet connectivity and damaged furniture. These requests need to be recorded, assigned to suitable maintenance personnel and followed up until the work is completed.

When this information is maintained in separate registers or spreadsheets, administrators may need to search multiple records to identify a student, room or responsible staff member. Tracking the progress of each complaint and preparing category-wise summaries can also take additional time. A centralised application can make these records easier to access and review.

Hostels require an organised process for maintaining student details, staff information, room allocations and facility-related complaints. These records support daily administration and help ensure that maintenance issues are assigned and followed up.

The Hostel Maintenance Management System uses Zoho Creator forms, list reports and dashboard components to organise hostel records. It provides a common interface for student, staff, room, complaint and feedback information. Deluge examples are documented as proposed validation and date-handling enhancements; they are not described as active workflows because their configuration was not verified in the supplied screenshots.

1.2 Problem Statement

When hostel information and maintenance requests are maintained manually or in disconnected files, administrators may face difficulty retrieving student and room details, identifying pending complaints, checking staff availability and monitoring resolution. A centralised system is required to organise these records and provide clear complaint status views.

- Scattered records
- Difficulty tracking pending work

- Time-consuming complaint follow-up

1.3 Objectives

- To develop a centralised hostel maintenance application using Zoho Creator.
- To capture and maintain student details including register number, department, year, hostel block and room number.
- To maintain staff contact details, staff categories and availability.
- To record hostel room capacity, occupied count, room status and assigned students.
- To capture maintenance complaints with category, description, priority, date and supporting photo.
- To provide complaint status views, feedback records and a dashboard for administrative monitoring.

1.4 Scope of the Project

In-scope:

- Maintaining student and staff information.
- Recording hostel blocks, floors, room numbers, capacity and occupancy.
- Capturing maintenance complaints and viewing complaint history.
- Reviewing all, pending, in-progress and resolved complaints.
- Displaying complaint statistics and category-wise issues.
- Recording complaint feedback, ratings and comments.
- Maintaining staff assignment and availability information.
- Reviewing resolution notes and dates where applicable.

Out-of-scope:

- Automated email/SMS notifications are not evidenced in the supplied screenshots.
- External payment processing is not part of the demonstrated application.
- Integration with external hostel or student information systems is not demonstrated.

- Predictive maintenance and advanced analytics are outside the demonstrated scope.

1.5 Justification for a Low-Code Approach

Zoho Creator is suitable for this project because it provides a low-code environment for configuring data-entry forms, reports, navigation and dashboard components. The visual form builder supports structured capture of student, staff, room, complaint and feedback information. Built-in reports help administrators review saved records without creating a separate interface for each view.

The platform provides a common environment for organising application data and presenting it through list views and dashboard components. The supplied screenshots show the application in the Development environment. No custom Deluge scripts, external integrations or notification workflows are claimed because they are not evidenced by the supplied screens.

Compared with maintaining separate manual registers, the proposed low-code application provides a structured and consolidated way to enter and review hostel maintenance information. It is appropriate for demonstrating form configuration, reports and dashboard-based monitoring.

Reference: Zoho Corporation, Zoho Creator Documentation. Available at: https://www.zoho.com/creator/help

1.6 Organisation of the Report

Chapter 1 introduces the project, its objectives and scope. Chapter 2 studies the existing process and proposed system. Chapter 3 presents the requirements. Chapter 4 explains the system design. Chapter 5 describes the application modules and screenshots. Chapter 6 discusses testing. Chapter 7 presents deployment and security considerations. Chapter 8 discusses the observed results, and Chapter 9 concludes the report.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a conventional hostel maintenance process, student and room information may be stored in registers or spreadsheets. A student reports an issue to the hostel office, where the complaint is recorded and passed to a maintenance worker. Administrators then follow up to determine whether the issue has been attended to and record the outcome.

The process involves identifying the student and room, classifying the issue, assigning appropriate maintenance staff, tracking the status and recording the resolution. As the number of residents and complaints increases, manual tracking can become difficult and category-wise summaries require additional effort.

The proposed application organises the maintenance process through dedicated Zoho Creator forms, list reports, status views and an administrative dashboard.

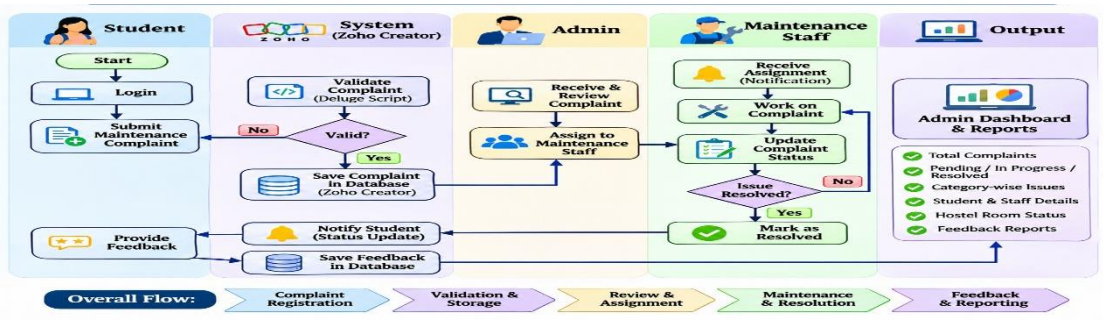


Figure 2.1: As-Is Process Flow of the Existing System 4

2.2 Limitations of the Existing System

- Manual calculations
- Lack of automation
- Time-consuming reports

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application 4

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development approach	Requires application code and database configuration	Visual forms, reports and page components

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Skills required	Programming and database knowledge	Platform configuration and basic scripting knowledge
Data storage	Database must be selected and configured	Application data is managed within Zoho Creator
Reports	Custom report interfaces and queries may be required	Built-in list and report views
Dashboard	Charts and KPI cards are custom-built	Dashboard components can be configured visually
Maintenance	Changes may require code modification and deployment	Many changes can be made through visual configuration
Customisation	High flexibility	Bounded by platform capabilities and plan

2.4 Literature Survey

Table 2.2: Summary of Literature Survey 5

S.No.	Author(s) / Source	Title / Product	Key Findings / Coverage	Relevance to this Project
1	Zoho Corporation	Zoho Creator Help Documentation	Explains forms, reports, pages and application configuration	Platform used to build the application
2	Zoho Corporation	Deluge Help Documentation	Describes scripting syntax and workflow actions	Reference for proposed validation scripts
3	OWASP Foundation	Application Security Verification Standard	Provides guidance on authentication, access control and data validation	Relevant to protecting student and staff records
4	Richardson (2014)	Low-Code Platforms	Discusses visual development and reduced coding effort	Supports the low-code approach
5	IEEE Transactions on Reliability (2024)	Empirical Study of Low-Code Platform Bugs	Examines defects and reliability considerations in low-code platforms	Highlights the need for testing and validation

2.5 Emerging Trends in Low-Code Development

Low-code development supports the creation of data-driven administrative applications. In a hostel maintenance context, visual forms support structured data capture, reports make records easier to review, and dashboard charts help administrators understand complaint status and issue categories.

1. **Visual Form Configuration: Forms can be configured to capture student, staff, room, complaint and feedback details.**
2. **Report Views: Separate views help administrators review all complaints, pending complaints, resolved complaints and category-wise records.**
3. **Centralised Records: Student, staff and room information can be organised within one application.**

4. **Data Visualisation: Dashboard cards and charts provide a summary of complaint status and category distribution.**
5. **Access Management: Application permissions should be configured so that only authorised users can access sensitive records.**

These capabilities support a structured approach to hostel maintenance administration. The features described as implemented in this report are limited to those visible in the supplied application screenshots.

2.6 Proposed System

The proposed Hostel Maintenance Management System is a low-code application developed using Zoho Creator. It provides a centralised interface for student details, staff details, hostel rooms, maintenance complaints and complaint feedback.

The application provides forms for student, staff, hostel room, maintenance complaint and complaint feedback information. List reports provide access to saved records. Separate complaint views support monitoring by status, staff assignment, category and student history, while the dashboard presents summary cards and charts.

The proposed system addresses the limitations of manual record keeping by organising hostel information in a single application and making complaint status and category summaries easier to review.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles 7

Role	Description	Key Responsibilities in the Application
Administrator	Hostel administration / application owner	Maintain student, staff and room records; review complaints; monitor reports and dashboard
Student	Hostel resident / complaint reporter	Provide complaint details and submit feedback; student-facing portal access is not evidenced
Maintenance Staff	Personnel who attend maintenance work	Review assigned work and update complaint progress where authorised; separate staff login is not evidenced

3.2 Functional Requirements

Table 3.2: Functional Requirements 7

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall capture and maintain student register number, contact, department, year and hostel allocation details.	Administrator	High
FR-02	The system shall capture staff name, phone, email, staff category and availability.	Administrator	High
FR-03	The system shall maintain hostel block, floor, room number, capacity, occupied count and room status.	Administrator	High
FR-04	The system shall record maintenance complaints with student, room, category, description, priority and date.	Administrator / authorised user	High
FR-05	The system shall provide views for all, pending and resolved complaints and complaint history.	Administrator	High
FR-06	The system shall provide staff assignment and category-wise complaint views.	Administrator	Medium
FR-07	The system shall capture complaint feedback, rating, comments and feedback status.	Administrator / authorised user	Medium
FR-08	The dashboard shall display complaint totals, status summary and category-wise issue counts.	Administrator	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements 7

NFR ID	Category	Requirement
NFR-01	Usability	Forms and navigation should be understandable to hostel administrative users.
NFR-02	Security	Access to student, staff, room and complaint records should be restricted to authorised users.
NFR-03	Availability	The application should be accessible when Zoho Creator and the internet connection are available.
NFR-04	Data integrity	Required fields and suitable field types should reduce incomplete or inconsistent entries.
NFR-05	Maintainability	Forms, reports and dashboard components should be organised for future updates.
NFR-06	Performance	List and dashboard views should remain responsive for the expected hostel record volume.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements 8

Category	Specification
Platform	Zoho Creator (Development environment shown in screenshots)
Low-code scripting	Deluge examples proposed for form validation and date handling; execution not verified
Browser	Recent version of Google Chrome, Microsoft Edge or Mozilla Firefox
Client device	Desktop or laptop computer
Connectivity	Internet connection
Application components	Zoho Creator forms, list reports, navigation and dashboard components
External API tools	Not required for the demonstrated application; no external API is evidenced

3.5 Use Case Model

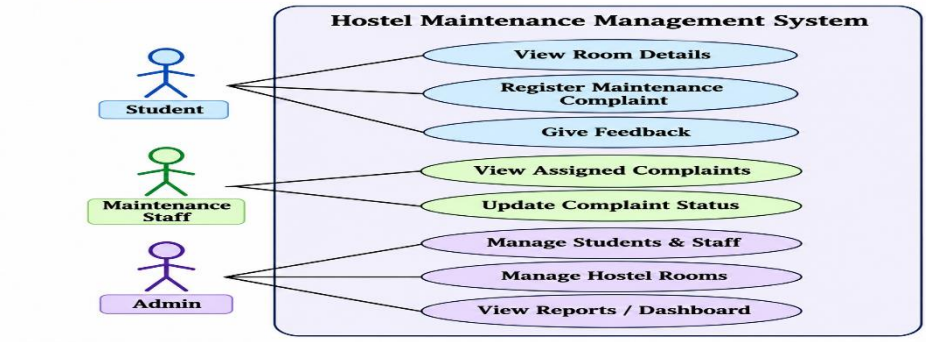


Figure 3.1: Use Case Diagram of the Hostel Maintenance Management System 8

Table 3.5: Use Case Description – Add Complaint (UC-01) 8

Use Case ID	UC-01
Use Case Name	Register Maintenance Complaint
Actor(s)	Administrator / authorised user; student is the complaint source
Pre-condition	Application is accessible and relevant student / room information is available.
Main Flow	1. Open Maintenance Complaints. 2. Enter student, block, room, category, description and priority. 3. Submit the form. 4. Review the saved complaint in the list view.
Alternate Flow	Missing required details or invalid values should be corrected before submission; validation rules must be configured and tested.
Post-condition	Complaint record is saved and can be reviewed in the complaint views.
Related FR	FR-04, FR-05

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

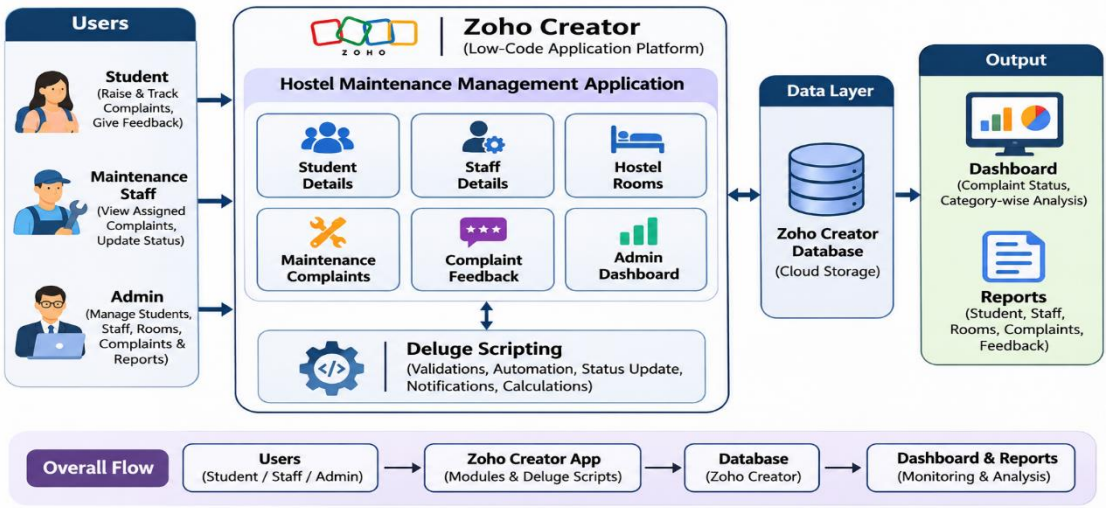


Figure 4.1: System Architecture of the Hostel Maintenance Management System 9

User Layer: The hostel administrator and authorised users access the application through the Zoho Creator interface. Student and maintenance-staff activities are represented as application use cases; separate portal access is not evidenced.

Application Layer: Zoho Creator provides the navigation menu, data-entry forms, list reports and dashboard components.

Process Layer: Users submit student, staff, room, maintenance complaint and feedback information through the configured forms. Complaint records can be reviewed through the available status and category views. Proposed Deluge scripts describe validation and date-handling logic.

Data Layer: The application organises records captured through Student Details, Staff Details, Hostel Rooms, Maintenance Complaints and Complaint Feedback.

Output Layer: List views display saved records, while the Hostel Admin Dashboard presents complaint totals, status distribution and category-wise maintenance issues.

4.2 Data Model

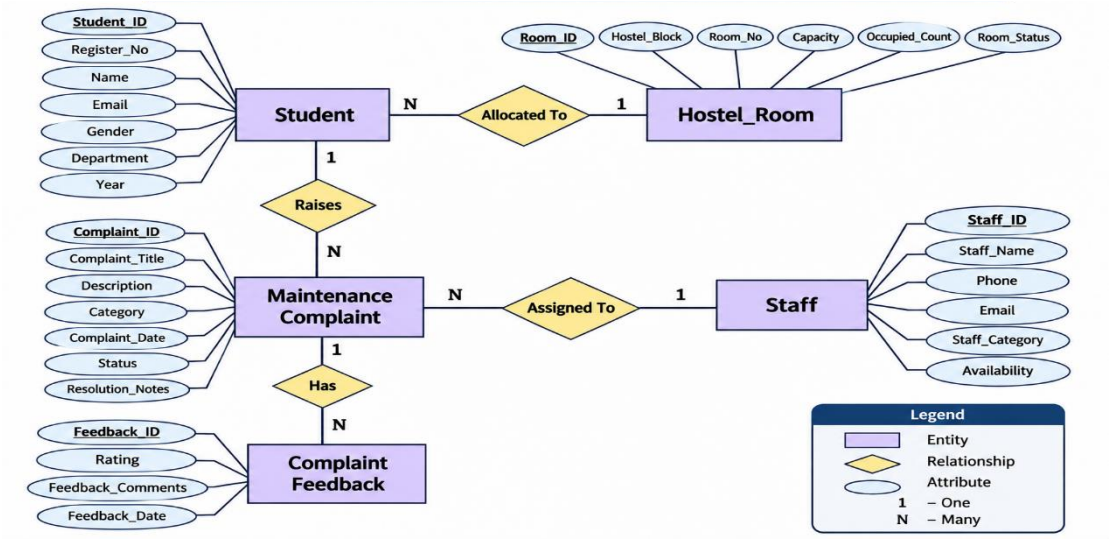


Figure 4.2: Entity–Relationship Diagram of the Hostel Maintenance Management System 10

Table 4.1: Entities and Relationships 10

Entity (Form)	Key Fields	Related To	Relationship Type	Evidence / Notes
Student Details	Register Number, Email, Gender, Department, Year, Hostel Block, Room Number, Phone	Hostel Rooms; Maintenance Complaints; Complaint Feedback	Logical association / reference	Student form and list view shown; exact lookup configuration not visible
Hostel Rooms	Hostel Block, Floor Number, Room Number, Room Capacity, Occupied Count, Room Status, Assigned Student	Student Details; Maintenance Complaints	Logical association / reference	Room form and 11-row list shown
Staff Details	Staff Name, Phone, Email, Staff Category, Availability	Maintenance Complaints	Logical association / reference	Staff form and list view shown; exact assignment lookup not visible
Maintenance Complaints	Student, Hostel Block, Room Number, Complaint Category, Description, Priority, Complaint Date, Assigned Staff, Status, Resolution Notes, Resolved Date	Student Details; Hostel Rooms; Staff Details; Complaint Feedback	Logical association / reference	Complaint form and status/category views shown
Complaint Feedback	Complaint, Student, Rating, Feedback Comments, Feedback Date, Feedback Status, Staff Responder, Resolution Time, Suggested Improvement	Maintenance Complaints; Student Details; Staff Details	Logical association / reference	Feedback form and one sample feedback record shown

4.3 Form Design

Table 4.2: Field Specification – Student Details Form 10

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Register Number	Not visible	Single Line	Yes	Required field (marked *)	Student identifier
Email	Not visible	Email	No	Email format	Student contact
Gender	Not visible	Radio	No	Male / Female / Others	Gender selection
Department	Not visible	Dropdown	No	Select department	Academic department
Year	Not visible	Dropdown	No	Select year	Year of study

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Hostel Block	Not visible	Dropdown	No	Select block	Hostel allocation
Room Number	Not visible	Single Line	Yes	Required field (marked *)	Room allocation
Phone	Not visible	Phone	Yes	Country code +91 shown	Contact number

Table 4.3: Field Specification – Maintenance Complaint Form 11

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student	Not visible	Lookup / text	No	Select student	Complaint owner
Hostel Block	Not visible	Dropdown	No	Select block	Hostel location
Room Number	Not visible	Single Line	Yes	Required field (marked *)	Room identification
Complaint Category	Not visible	Dropdown	Yes	Select category	Issue classification
Complaint Title	Not visible	Single Line	No	Optional in screenshot	Short issue title
Description	Not visible	Multi Line	Yes	Required field (marked *)	Problem details
Complaint Photo	Not visible	Image upload	No	Select image	Supporting evidence
Priority	Not visible	Dropdown	Yes	Select priority	Urgency classification
Complaint Date	Not visible	Date-time	No	Date/time placeholder shown	Submission date
Assigned_Staff	Not visible	Lookup / text	No	Staff selection	Assigned maintenance worker
Student details	Not visible	Dropdown / lookup	Yes	Select student details	Student reference field
Staff Details	Not visible	Dropdown / lookup	No	Select staff details	Staff reference field
Status	Not visible	Dropdown	No	Submitted shown as default	Complaint progress
Resolution_Notes	Not visible	Multi Line	No	Optional	Work completed / resolution notes
Resolved_Date	Not visible	Date	No	Optional	Date of resolution

4.4 Process and Workflow Design

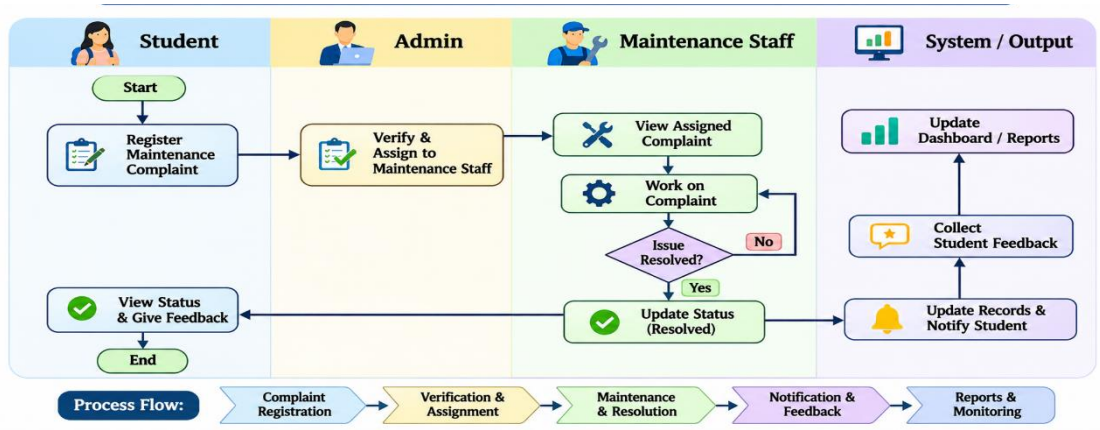


Figure 4.3: Hostel Maintenance Complaint Workflow 11

4.5 User Interface and Navigation Design

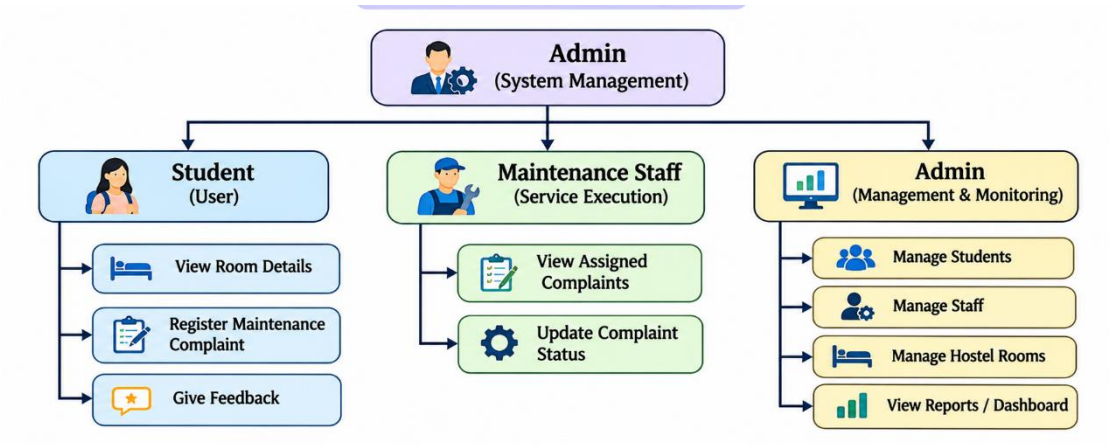


Figure 4.4: Hostel Maintenance Application Hierarchy / Navigation 11

4.6 Access Control Design

The screenshots do not confirm separate role profiles or record-level permissions. The matrix therefore identifies the access controls that should be configured and verified before live use; it does not claim that these permissions are already active.

Table 4.4: Role–Permission Matrix 12

Component	Administrator	Student / Standard User	Maintenance Staff	Notes
Student Details form	To configure	To verify	Not applicable	Permission settings are not visible in screenshots
Staff Details form	To configure	Not applicable	To verify	Restrict staff directory editing to authorised users
Hostel Rooms form	To configure	View access if required	View access if required	Portal and record-level access not evidenced
Maintenance Complaint form	To configure	Submit / view own complaints if enabled	View assigned complaints if enabled	Actual permissions must be checked in Zoho Creator
Complaint Feedback form	To configure	Submit feedback if enabled	Not applicable	Student portal access is not evidenced
Dashboard and reports	To configure	Limited view if enabled	Assigned-work view if enabled	No role-specific profiles are claimed as implemented

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The Hostel Maintenance Management System was configured in Zoho Creator as a low-code application. The visible navigation contains Student Details, Staff Details, Hostel Rooms, Maintenance Complaints, Complaint Feedback and Hostel Admin Dashboard. The application screenshots show the Development environment.

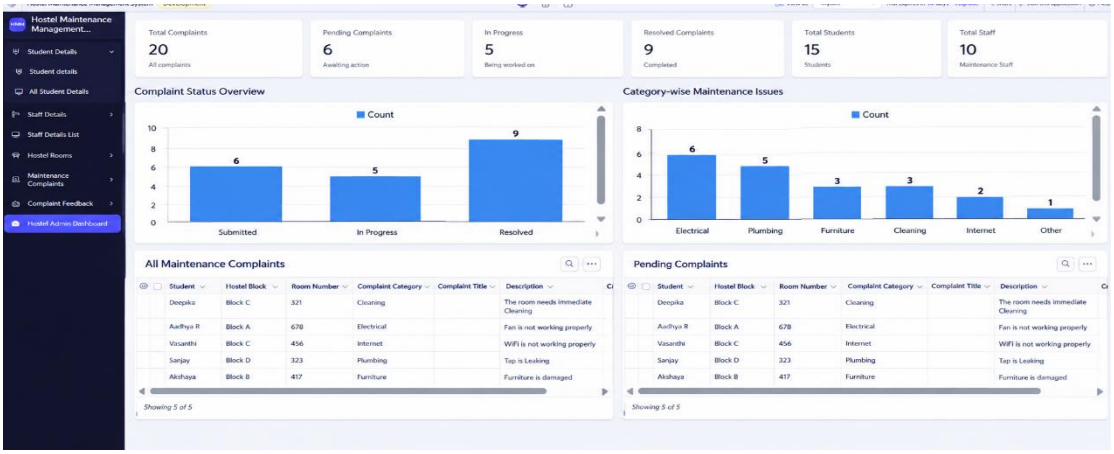


Figure 5.1: Hostel Admin Dashboard and Application Navigation 14

5.2 Forms and Data Capture

The application contains dedicated forms and list views for student, staff, hostel room, maintenance complaint and complaint feedback information. Student Details captures register number, email, gender, department, year, hostel block, room number and phone. Staff Details captures name, phone, email, staff category and availability. Hostel Rooms captures block, floor, room number, capacity, occupied count, status, assigned student and inspection-related information.

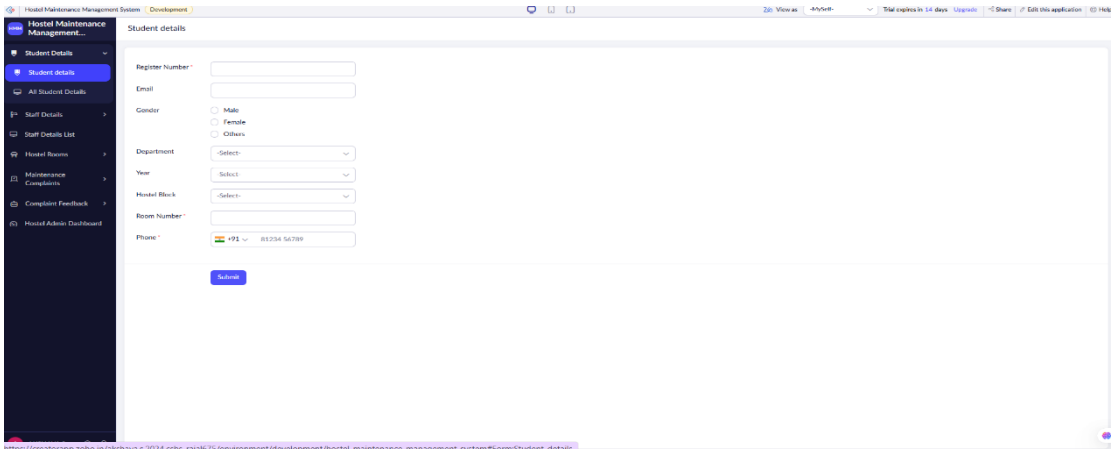


Figure 5.2: Student Details Form 15

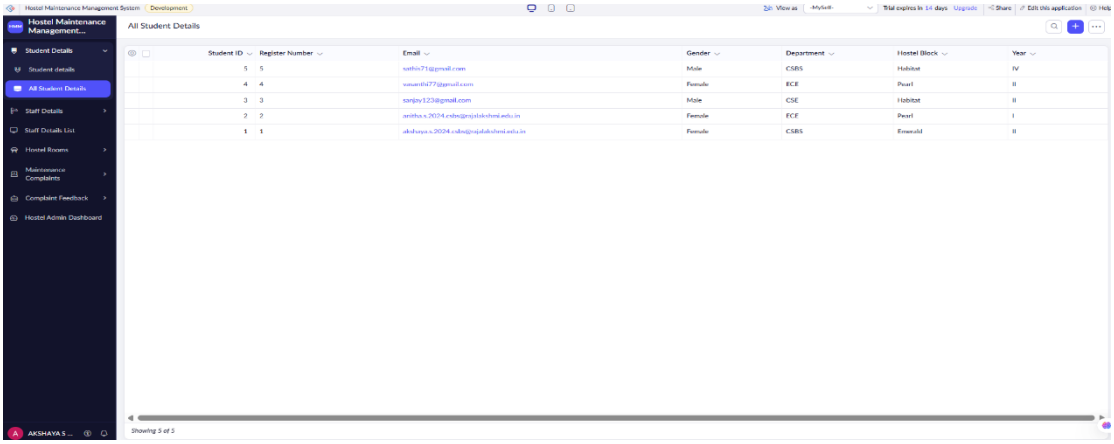


Figure 5.3: All Student Details List 15

5.3 Reports

Table 5.1: Reports Implemented 14

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Student Details	List report	Student Details	Displays register number, email, gender, department, block and year	Search / standard report controls	Application user
Staff Details List	List report	Staff Details	Displays staff name, phone, email, category and availability	Search / standard report controls	Application user
All Hostel Rooms	List report	Hostel Rooms	Displays block, floor, room, capacity, occupancy, status and assigned student	Add record / search	Application user
All Maintenance Complaints	List report	Maintenance Complaints	Displays complaint records and categories	Search / standard report controls	Application user
All Complaints	List report	Maintenance Complaints	Displays all complaint records	Search / standard report controls	Application user
Pending Complaints	Filtered report	Maintenance Complaints	Shows complaints awaiting action	Status-based view	Application user
Resolved Complaints	Filtered report	Maintenance Complaints	Shows completed complaints	Status-based view	Application user

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Staff Assignments	List / filtered report	Maintenance Complaints and Staff Details	Supports review of staff assignments	Search / standard report controls	Application user
Category Wise Complaints	Grouped report	Maintenance Complaints	Groups records by complaint category	Category grouping	Application user
Student Complaint History	Filtered report	Maintenance Complaints and Student Details	Shows complaint history by student	Student-based view	Application user
All Complaint Feedback	List report	Complaint Feedback	Displays feedback, rating, comments and status	Search / standard report controls	Application user
Feedback Report	Report	Complaint Feedback	Provides feedback review	Standard report controls	Application user
Hostel Admin Dashboard	Dashboard	Student, Staff, Hostel Rooms and Maintenance Complaints	Summary cards and status/category charts	Dashboard components	Application user

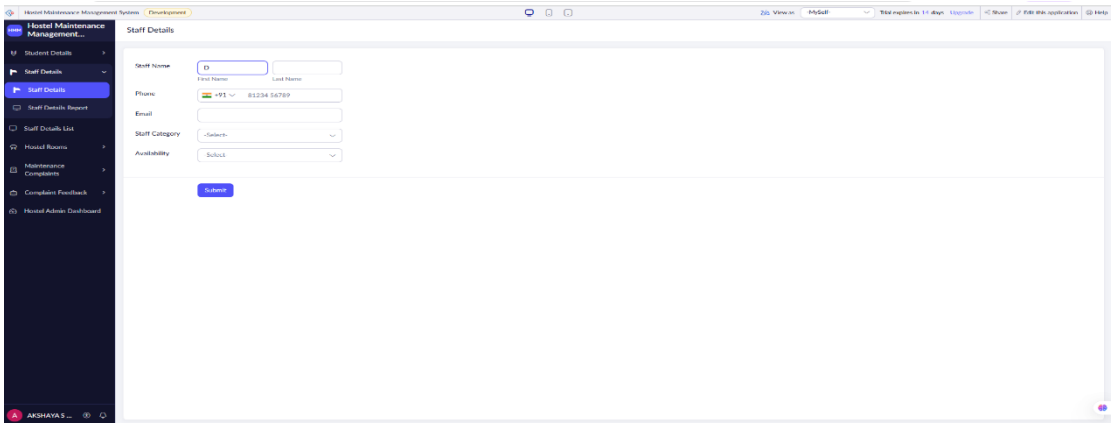


Figure 5.4: Staff Details Form 16

5.4 Pages and Dashboards

The Hostel Admin Dashboard presents summary cards for Total Complaints, Pending Complaints, In Progress, Resolved Complaints, Total Students and Total Staff. The supplied screenshot shows 20 complaints, 6 pending, 5 in progress, 9 resolved, 15 students and 10 staff. The category-wise chart shows Electrical 6, Plumbing 5, Furniture 3, Cleaning 3, Internet 2 and Other 1. Both the category counts and status counts add up to 20.

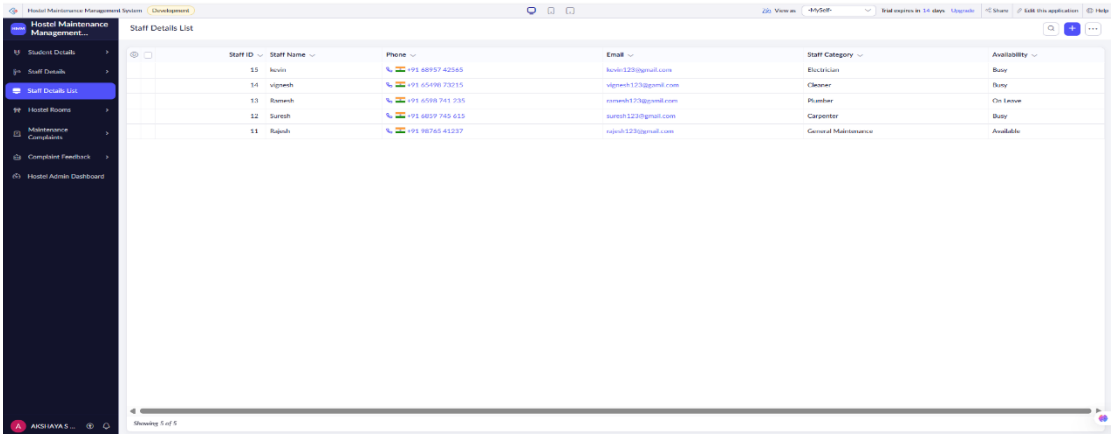


Figure 5.5: Staff Details List 17

5.5 Complaint Tracking and Status Views

Table 5.2: Proposed Deluge Validation Scripts 17

Script ID	Location / Trigger	Purpose	Implementation Evidence
DS-01	Hostel Rooms – On Validate	Validate room capacity and occupied count.	Proposed Deluge example in Section 5.5.2; configuration and execution not verified.
DS-02	Maintenance Complaints – On Validate	Set complaint date and initial Submitted status.	Proposed Deluge example in Section 5.5.3; configuration and execution not verified.
DS-03	Maintenance Complaints – On Validate	Populate or clear resolved date based on status.	Proposed Deluge example in Section 5.5.4; configuration and execution not verified.
DS-04	Complaint Feedback – On Validate	Validate rating range from 1 to 5.	Proposed Deluge example in Section 5.5.5; configuration and execution not verified.

The supplied screenshots demonstrate the configured forms, reports and dashboard. The Deluge examples in this report are proposed scripts for extending the application; they have not been verified as configured or executed in the supplied Zoho Creator application.

The application navigation includes All Maintenance Complaints, All Complaints, Pending Complaints, Resolved Complaints, Staff Assignments, Category Wise Complaints and Student Complaint History. The Complaint Feedback section includes the feedback form, All Complaint Feedback and Feedback Report.

The dashboard provides a consolidated view of complaint status and category distribution. No email/SMS notification workflow or external integration is evidenced in the supplied screenshots.

5.5.1 Deluge Scripting for Validation and Automation

Deluge is the scripting language available in Zoho Creator. It can be used with form workflows to validate entries and apply business rules. The following examples are proposed implementation scripts. They must be added to the corresponding form workflows and tested after the actual field link names and field types are confirmed.

Important: The exact form and field link names are not visible in the supplied screenshots. The examples below assume link names such as Room_Capacity, Occupied_Count, Complaint_Date, Status, Resolved_Date and Rating. Open each field's properties in Zoho Creator and replace these assumed names with the actual link names before using the scripts.

5.5.2 Script 1 – Validate Hostel Room Capacity

Location: Hostel Rooms form → On Validate workflow. Purpose: prevent negative capacity values and ensure that occupied count does not exceed room capacity.

```
if(input.Room_Capacity < 0 || input.Occupied_Count < 0)
{
    alert "Capacity and occupied count cannot be negative.";
    cancel submit;
}

if(input.Occupied_Count > input.Room_Capacity)
{
    alert "Occupied count cannot exceed room capacity.";
    cancel submit;
}
```

Expected result: the form is prevented from submitting when capacity or occupied count is negative, or when occupied count is greater than room capacity. Confirm that both fields are numeric before using this script.

5.5.3 Script 2 – Set Complaint Date and Initial Status

Location: Maintenance Complaints form → On Validate workflow. Purpose: populate the complaint date when it is empty and set the initial status to Submitted when no status has been selected.

```
if(input.Complaint_Date == null)
{
    input.Complaint_Date = zoho.currenttime;
}

if(input.Status == null || input.Status == "")
{
    input.Status = "Submitted";
}
```

Expected result: a newly submitted complaint receives a date/time and an initial Submitted status, provided the field link names and field types match the application.

5.5.4 Script 3 – Record the Resolved Date

Location: Maintenance Complaints form → On Validate workflow. Purpose: record the date when a complaint is marked Resolved. The Resolved_Date field should be configured as a Date field.

```
if(input.Status == "Resolved")
{
    if(input.Resolved_Date == null)
    {
        input.Resolved_Date = zoho.currentdate;
    }
}
else
{
    input.Resolved_Date = null;
}
```

Expected result: when the complaint status is Resolved, the resolved date is populated if it is blank. If the status is changed back to another value, the date is cleared. Adjust this rule if the institution needs to retain the original resolution date for audit purposes.

5.5.5 Script 4 – Validate Complaint Feedback Rating

Location: Complaint Feedback form → On Validate workflow. Purpose: ensure that a numeric rating is within the permitted range of 1 to 5. This example assumes Rating is a Number field.

```
if(input.Rating < 1 || input.Rating > 5)
{
    alert "Please enter a rating between 1 and 5.";
    cancel submit;
}
```

Expected result: feedback is submitted only when the rating is between 1 and 5. If the Rating field is a dropdown or choice field rather than a Number field, adapt the comparison to the configured choice values.

These scripts are proposed validation and date-handling examples. They do not demonstrate automatic staff allocation, notifications or dashboard calculations. Those features require separate workflow configuration and testing.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented 19

Workflow / View	Type	Trigger / Event	Condition	Action(s)	Evidence
Student Details	Form	Manual form submission	Student information entered	Save student record through form	Form and list screenshots
Staff Details	Form	Manual form submission	Staff information entered	Save staff record through form	Form and list screenshots
Hostel Rooms	Form	Manual form submission	Room information entered	Save room record through form	Form and list screenshots
Maintenance Complaints	Form	Manual form submission	Complaint information entered	Save complaint record through form	Form and list screenshots
Complaint Status Views	Reports	Administrator opens view	Status/category view selected	Review complaint records	Navigation and list screenshots
Complaint Feedback	Form	Manual form submission	Feedback details entered	Save feedback record through form	Feedback form and list screenshot
Hostel Admin Dashboard	Dashboard	Administrator opens dashboard	Summary data available	Display complaint cards and charts	Dashboard screenshot

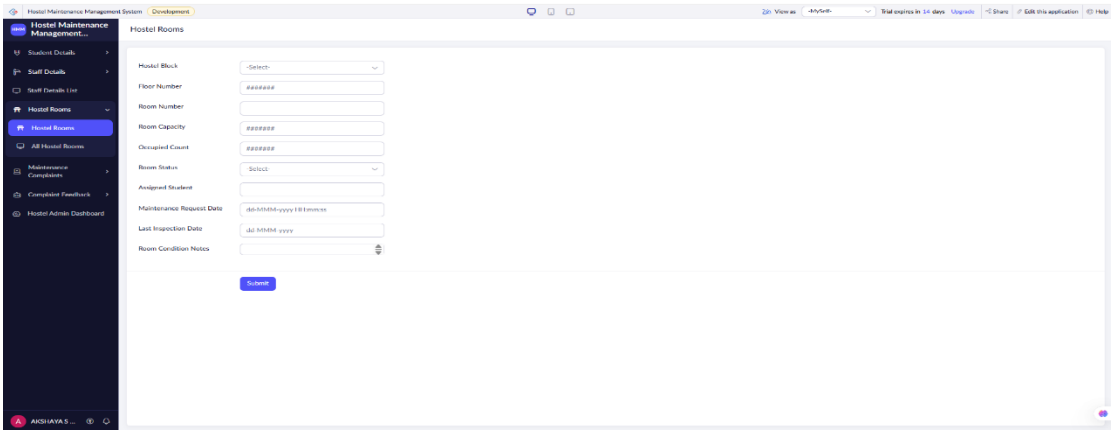


Figure 5.6: Hostel Rooms Form and Records 20

The supplied screenshots do not demonstrate an email notification received after complaint submission. Notifications may be considered as a future enhancement after workflow configuration and testing.

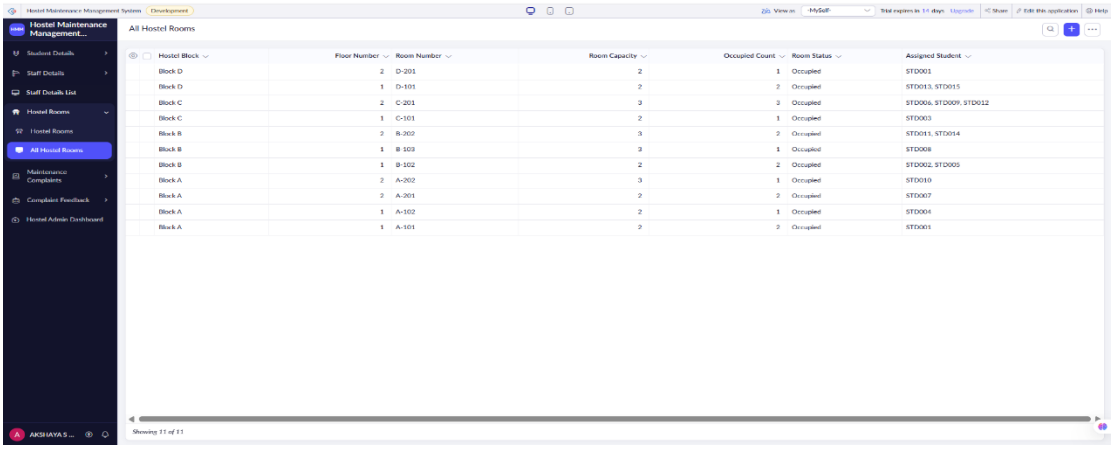


Figure 5.7: All Hostel Rooms Report 20

5.7 Integrations

Table 5.4: Integrations Implemented 20

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho Creator data store	Built-in application database	Forms and reports	Platform-managed	Stores student, staff, room, complaint and feedback records
External services	None evidenced	Not applicable	Not applicable	No external API, webhook or third-party integration is shown in the supplied screenshots.

No external API, REST call, webhook or third-party service is evidenced in the supplied screenshots. The demonstrated application uses Zoho Creator's built-in forms, reports and data storage.

Integration note: No external integration is configured in the demonstrated application.

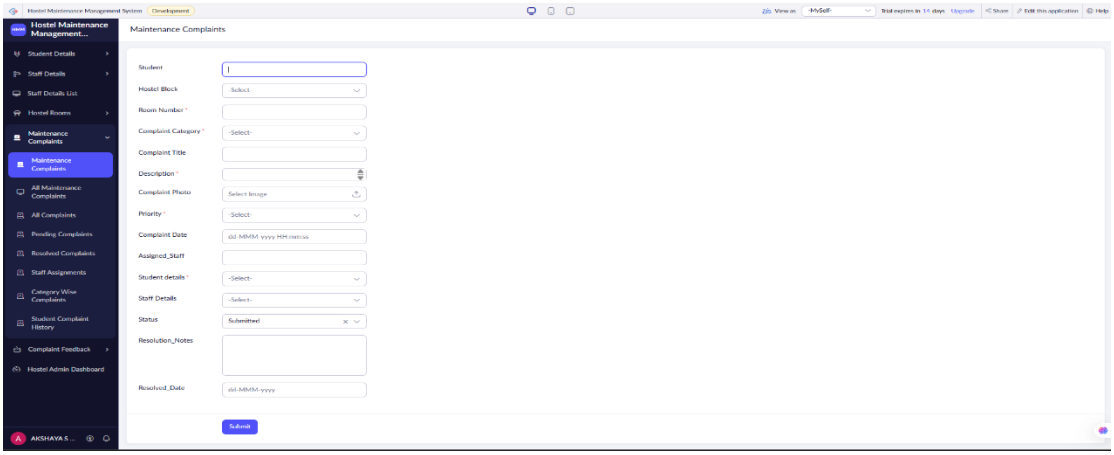


Figure 5.8: Maintenance Complaints Form 21

5.8 User Management and Role-Based Access

The screenshots show the signed-in application interface and its navigation sections. They do not establish that separate permission profiles, record-level rules or a student portal have been configured. Access control should be verified in the Zoho Creator application settings before live use.

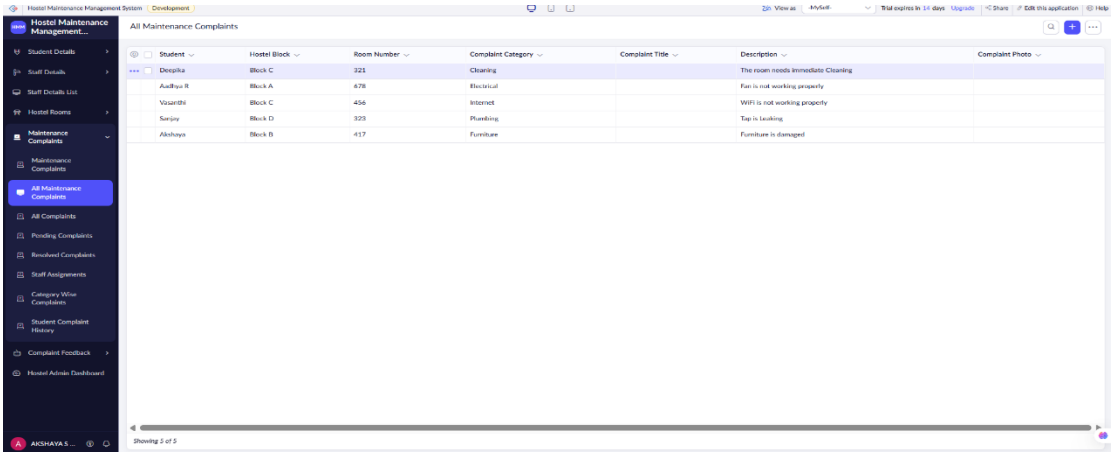


Figure 5.9: All Maintenance Complaints List 21

5.9 AI and Analytics Features

The Hostel Admin Dashboard shows summary cards and two charts: a complaint status overview and category-wise maintenance issues. The screenshot displays complaint counts by status and by category. No AI, predictive analytics or external analytics integration is evidenced.

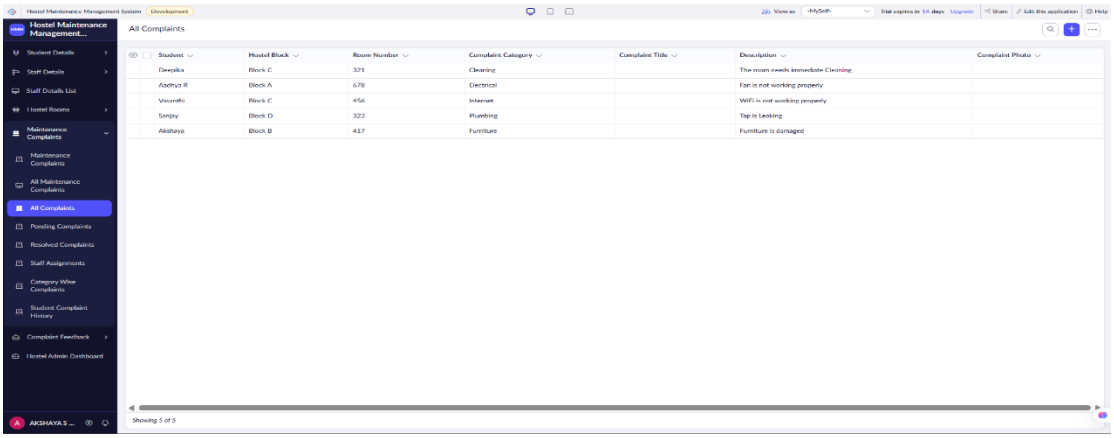


Figure 5.10: Category-wise Maintenance Complaints 22

5.10 Pro-Code Extension

The current project uses Zoho Creator forms, reports, navigation and dashboard components. No pro-code extension such as custom JavaScript widgets or external REST APIs is shown in the supplied evidence. The supplied screenshots demonstrate the application in desktop view. A separate mobile layout or student portal is not evidenced in the supplied materials and should be tested separately if required.

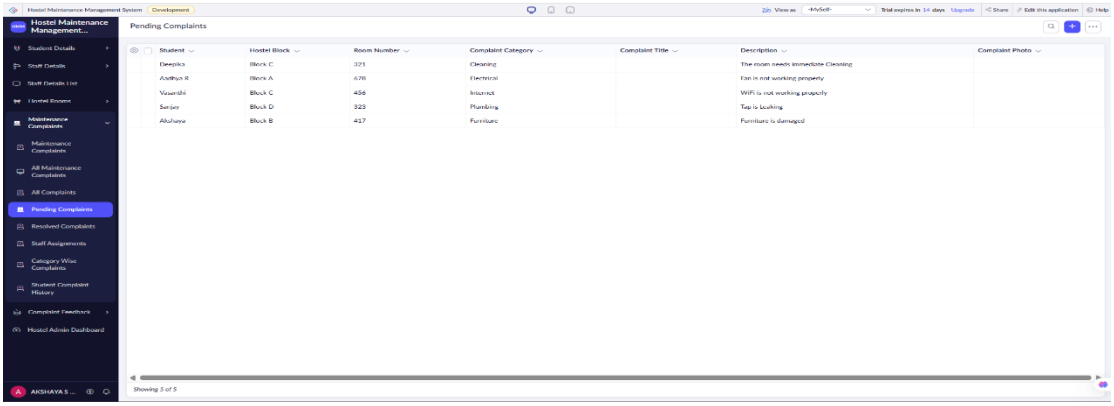


Figure 5.11: Student Complaint History View 22

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is organised around the visible application functions: opening each module, reviewing the form fields, checking the list views, reviewing complaint categories and comparing the dashboard totals with the displayed category and status counts. The supplied screenshots demonstrate configured screens and sample records, but do not constitute a complete automated test log.

6.2 Test Cases

Table 6.1: Test Cases 23

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result / Evidence	Status
TC-01	FR-01	Open Student Details	Select Student Details in navigation	Student form is displayed	Student Details form screenshot supplied	Evidence available
TC-02	FR-01	Review student records	Open All Student Details	Student list is displayed	5 sample rows visible	Evidence available
TC-03	FR-02	Open Staff Details	Select Staff Details in navigation	Staff form is displayed	Staff Details form screenshot supplied	Evidence available
TC-04	FR-02	Review staff records	Open Staff Details List	Staff list is displayed	5 sample rows visible	Evidence available
TC-05	FR-03	Open Hostel Rooms	Select Hostel Rooms form	Room fields are displayed	Hostel Rooms form screenshot supplied	Evidence available
TC-06	FR-03	Review room records	Open All Hostel Rooms	Room records are displayed	11 room records visible	Evidence available
TC-07	FR-04	Open complaint form	Select Maintenance Complaints	Complaint fields are displayed	Maintenance Complaints form screenshot supplied	Evidence available
TC-08	FR-04	Review complaint list	Open All Maintenance Complaints	Complaint records are displayed	5 sample complaint rows visible	Evidence available
TC-09	FR-05	Review status views	Open Pending and Resolved Complaints	Status-specific views are available	Both navigation items and resolved view are shown	Evidence available
TC-10	FR-06	Review category view	Open Category Wise Complaints	Records are grouped by category	Cleaning, Electrical, Furniture, Internet and Plumbing shown	Evidence available
TC-11	FR-07	Open feedback form	Select Complaint Feedback	Feedback fields are displayed	Complaint Feedback form screenshot supplied	Evidence available
TC-12	FR-07	Review feedback records	Open All Complaint Feedback	Feedback list is displayed	1 record with rating 4 and Reviewed status	Evidence available
TC-13	FR-08	Review dashboard totals	Open Hostel Admin Dashboard	Summary cards and charts are displayed	20 complaints; 6 pending; 5 in progress; 9 resolved; 15 students; 10 staff	Evidence available
TC-14	FR-08	Check category totals	Read category chart values	Category counts should total 20	6 + 5 + 3 + 3 + 2 + 1 = 20	Consistent with screenshot
TC-15	NFR-04	Validate room capacity	Try occupied count greater than room capacity using DS-01	Form should block invalid occupancy	Proposed Deluge example; not executed	Pending configuration
TC-16	FR-04	Set complaint date and initial status	Submit complaint with date/status blank using DS-02	Date and Submitted status should be populated	Proposed Deluge example; not executed	Pending configuration
TC-17	FR-05	Record resolved date	Set status to Resolved using DS-03	Resolved date should be populated	Proposed Deluge example; not executed	Pending configuration
TC-18	FR-07	Validate feedback rating	Enter rating outside 1–5 using DS-04	Form should block invalid rating	Proposed Deluge example; not executed	Pending configuration

6.3 Validation and Data Consistency

Debugging and validation should include checking mandatory fields, email and phone formats, room occupancy against room capacity, and consistency of complaint status values across the forms and reports. The proposed Deluge examples should be tested in the relevant form workflows after their field link names and field types have been verified.

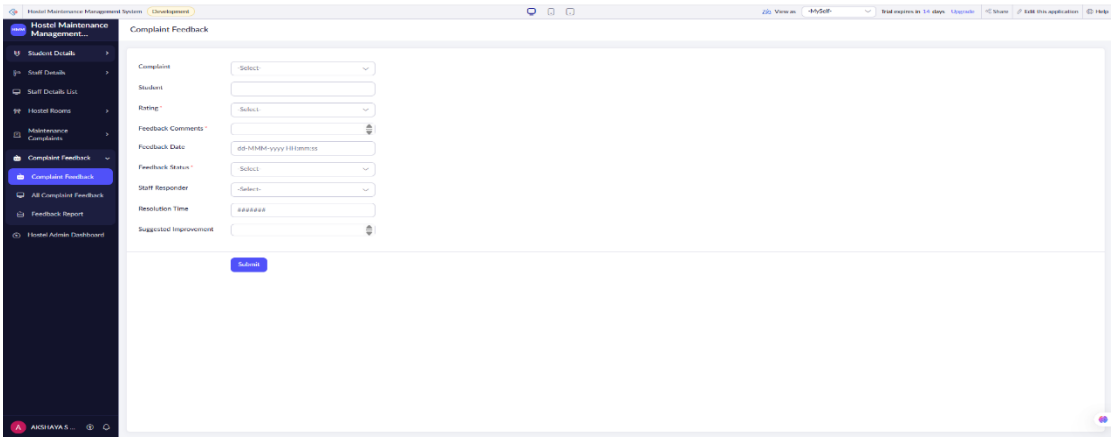


Figure 6.1: Complaint Feedback Form 25

6.4 Defect Log

Table 6.2: Defect Log 25

Issue ID	Observation	Evidence / Cause	Action / Recommendation	Status
OBS-01	Some sample complaint records have no complaint title visible.	The supplied list view shows blank title cells for sample rows.	Enter descriptive complaint titles when recording new complaints.	Recommendation
OBS-02	Room assignment and complaint room numbers should be consistent.	Room records and complaint records are shown in separate views.	Verify room identifiers and student allocations during data entry.	Recommendation
OBS-03	The screenshots do not confirm automatic notifications.	No notification workflow or received message is shown.	Configure and test notifications if required.	Future enhancement

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback 25

User	Role Tested	Feedback	Action Taken
Not recorded	Application user	Formal user feedback was not supplied.	Collect feedback during live user acceptance testing.
Not recorded	Application owner	Formal user feedback was not supplied.	Review the application with hostel administration users.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application is accessed through the Zoho Creator interface shown in the supplied screenshots. The evidence confirms that the application can be opened in the Development environment and its modules displayed, but it does not confirm a separate production deployment or a published application URL.

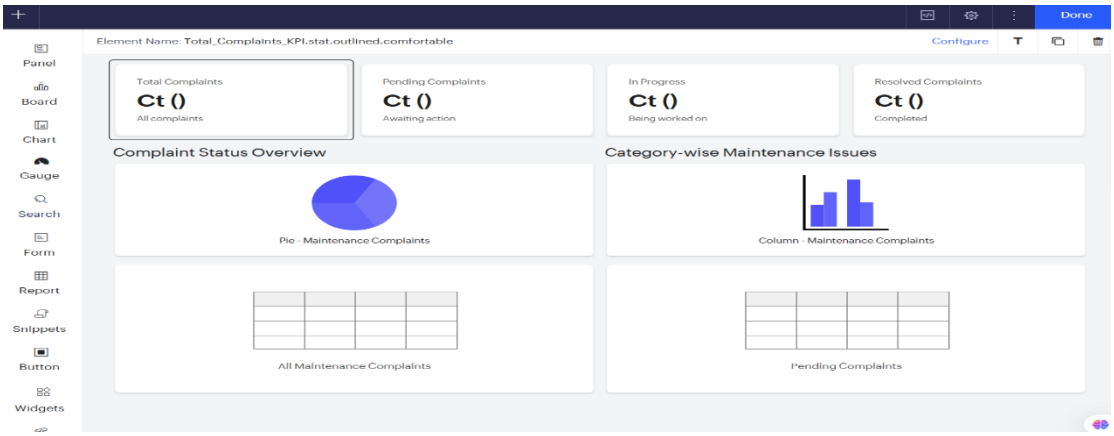


Figure 7.1: Hostel Admin Dashboard in the Development Environment **26**

Table 7.1: Deployment Checklist **26**

S.No.	Checklist Item	Evidence
1	Open all application modules	Navigation screenshots supplied
2	Review Student Details and Staff Details	Forms and list views supplied
3	Review Hostel Rooms	Form and 11-row list supplied
4	Review Maintenance Complaints	Form and complaint list supplied
5	Review status and category views	Pending, resolved and category views shown
6	Review Complaint Feedback	Feedback form and list supplied
7	Review Hostel Admin Dashboard	Summary cards and charts supplied
8	Verify production access and permissions	Not evidenced in supplied screenshots

7.2 Performance Optimisation

Table 7.2: Performance Optimisation Results 27

Optimisation / Design Practice	Component	Before	After	Observation
Use separate reports for different tasks	Student, staff and room reports	Not measured	Dedicated views shown	Improves navigation and record review
Group complaint views by status	Complaint reports	Not measured	Pending and resolved views shown	Supports status monitoring
Use summary cards and charts	Dashboard	Detailed review required	Summary shown on dashboard	Provides quick visual overview

7.3 Security Features

Table 7.3: Security Controls Implemented 27

Security Control	Where Applied / To Verify	Purpose
Zoho Creator account sign-in	Application access	Restrict access to signed-in users
Role/profile permissions	Must be verified in application settings	Restrict access to student, staff and complaint records
Data minimisation	Student and staff forms	Collect only information required for hostel administration

7.4 Governance and Administration

The application should be maintained by its owner. Recommended practices include keeping student and staff records accurate, reviewing room occupancy, monitoring pending complaints, updating complaint status after work is completed, reviewing feedback and restricting access to authorised users. These are recommendations; the supplied screenshots do not show the configured permission matrix.

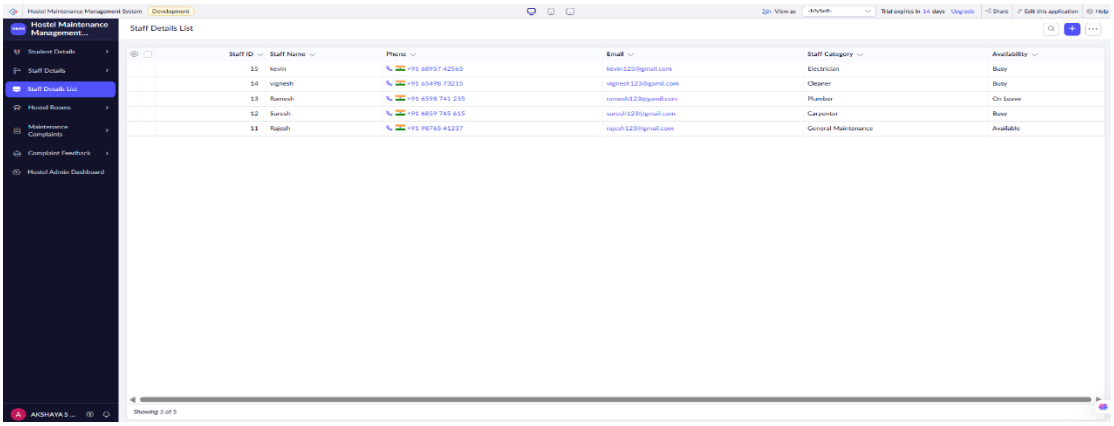


Figure 7.2: Staff Details List for Availability Review 30

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The Hostel Admin Dashboard provides a consolidated view of maintenance activity. The supplied dashboard displays 20 Total Complaints, 6 Pending Complaints, 5 In Progress, 9 Resolved Complaints, 15 Total Students and 10 Total Staff.

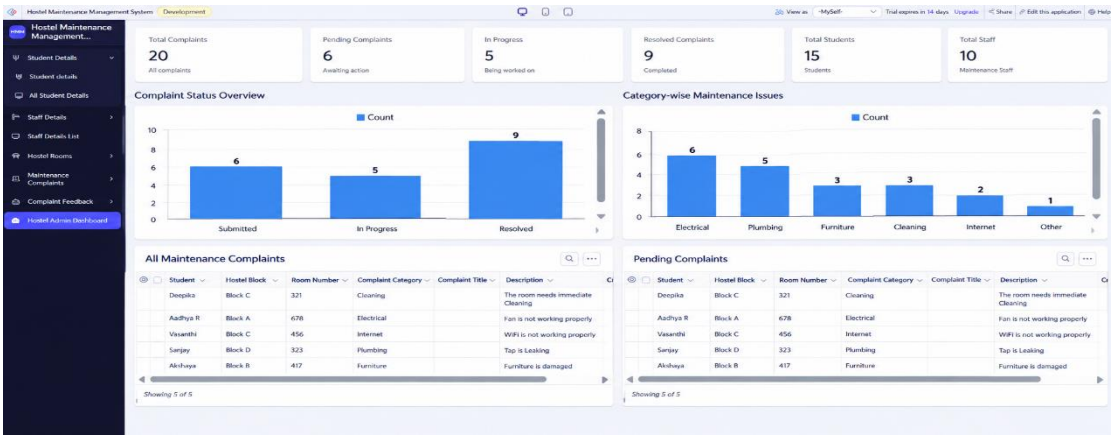


Figure 8.1: Hostel Admin Dashboard showing complaint status and category counts 31

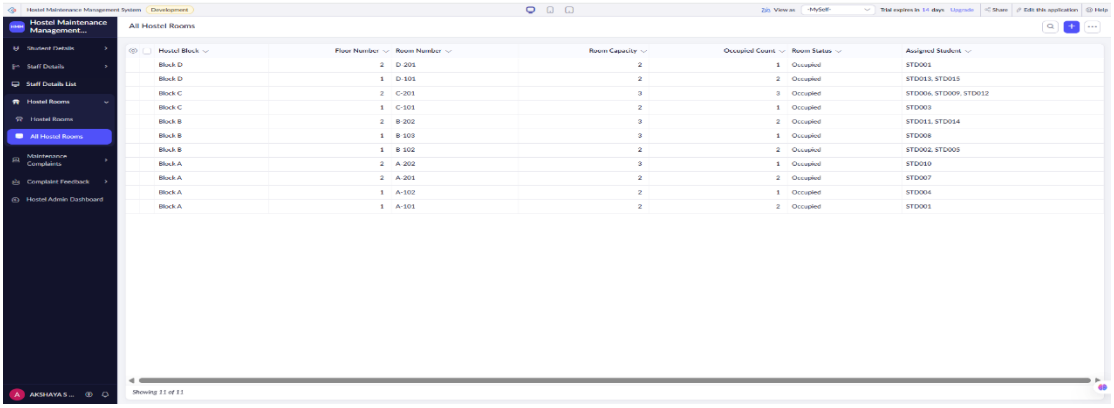


Figure 8.2: All Hostel Rooms list view 31

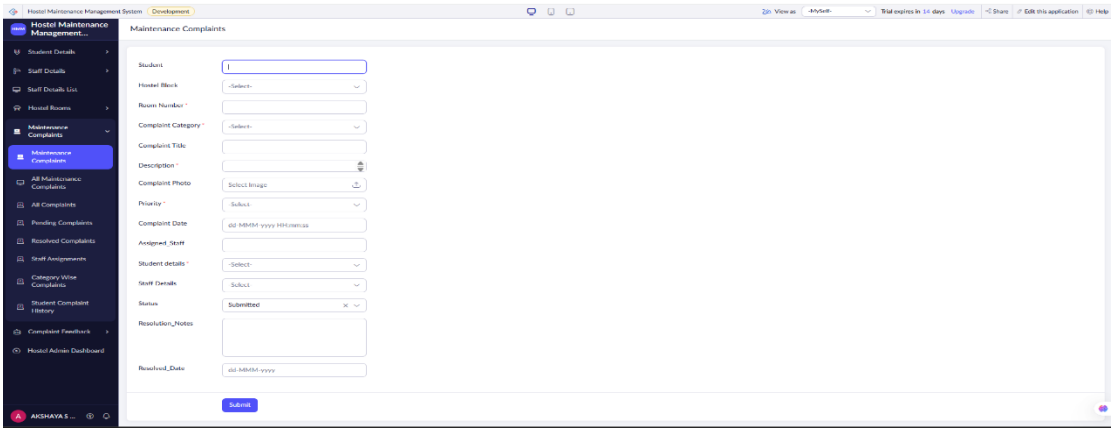


Figure 8.3: Maintenance Complaints form 32

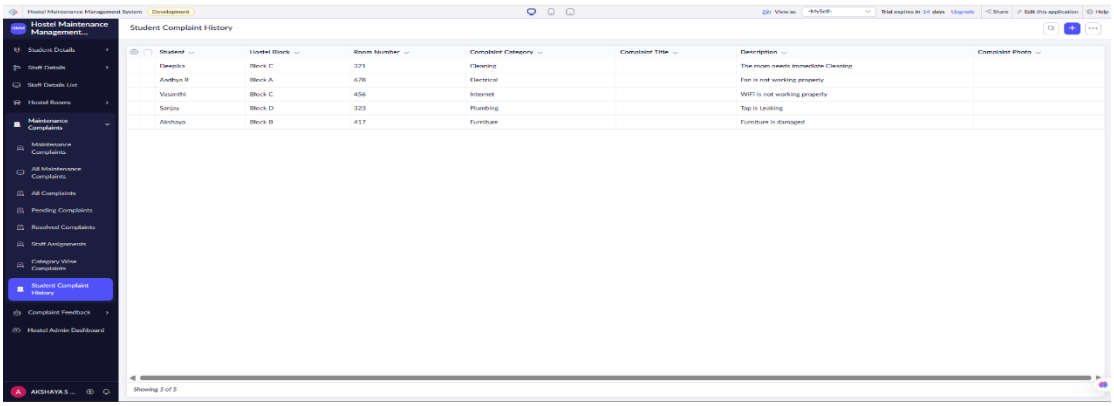


Figure 8.4: All Complaint Feedback list view 32

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes 32

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Maintain student information	Student Details form and All Student Details	5 sample student records visible
Maintain staff information	Staff Details form and Staff Details List	5 sample staff records visible
Maintain hostel room information	Hostel Rooms form and All Hostel Rooms	11 room records visible
Record maintenance complaints	Maintenance Complaints form and list views	5 sample complaint records visible

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Capture student feedback	Complaint Feedback form and list	1 feedback record visible
Monitor maintenance status	Hostel Admin Dashboard	20 complaints with status/category charts

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application 32

Parameter	Existing System	Proposed Application	Improvement
Student information	Manual register or spreadsheet	Zoho Creator Student Details form	Centralised digital records
Staff information	Separate staff register	Staff Details form and list	Staff category and availability visible
Room information	Manual room register	Hostel Rooms form and list	Room capacity and occupancy organised
Complaint tracking	Manual follow-up	Complaint form and status views	Pending and resolved records can be reviewed
Maintenance summary	Manual counting	Dashboard cards and charts	Quick visual summary

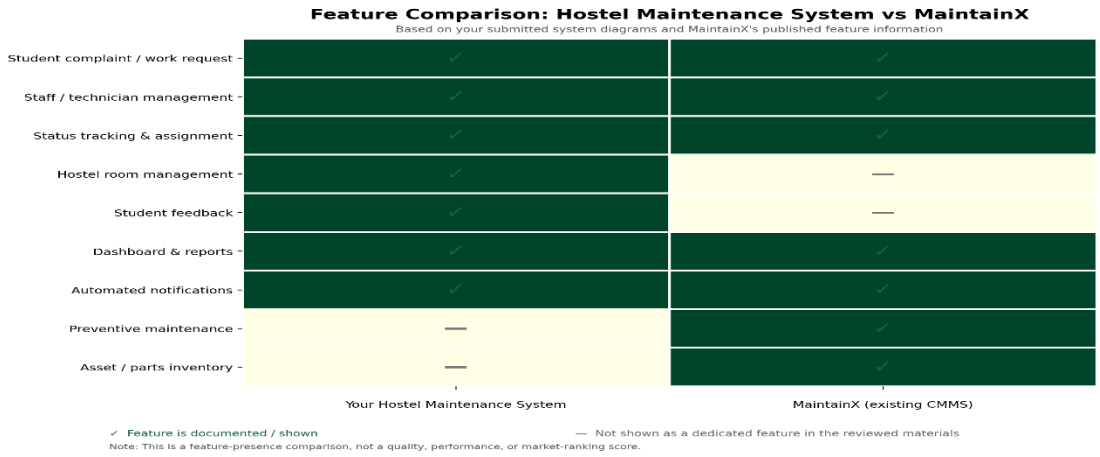


Figure 8.5: Existing System versus Proposed System – Qualitative Feature Comparison 33
Note: This figure is a qualitative comparison of feature availability. It is not a measured performance benchmark.

8.4 Limitations of the Low-Code Approach

The application depends on Zoho Creator's platform, data model and available configuration options. Visual forms, reports and dashboard components support rapid

development, while advanced workflow automation and integrations require additional configuration. The current report distinguishes visible implemented features from proposed enhancements.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The demonstrated application uses Zoho Creator's visual forms, reports, navigation and dashboard components. Proposed Deluge scripts are included as examples of low-code workflow extensions. The supplied screenshots do not verify that these scripts have been configured or executed.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Hostel Maintenance Management System was developed using Zoho Creator to organise student records, staff details, hostel room information, maintenance complaints and feedback in one application. The available list views support record review, while the dashboard provides a consolidated view of complaint status and category distribution. The supplied screenshot shows 20 complaints, 15 students and 10 staff.

9.2 Future Enhancements

1. Add notifications when a new maintenance complaint is submitted or when its status changes.
2. Configure staff assignment based on complaint category and staff availability, and add validation to prevent room occupancy from exceeding room capacity.
3. Add date filters and downloadable category-wise or category-wise maintenance reports; consider integrations with other hostel systems only if required.

9.3 Learning Outcomes and Career Relevance

This project provides practical experience in configuring forms, organising related records, creating reports and presenting operational information through a dashboard using a low-code platform. It also highlights the importance of field validation, data consistency, testing and access control.

REFERENCES

[1] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide. New Delhi, India: BPB Publications.

[2] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: Zoho Creator Help Documentation (Accessed: 31 October 2026).

[3] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/ (Accessed: 1 November 2026).

[4] Zoho Corporation, “Zoho Creator Learning Centre.” [Online]. Available: Zoho Creator Learning Centre (Accessed: 31 October 2026).

[5] Gartner, Inc., “Magic Quadrant for Enterprise Low-Code Application Platforms,” Gartner Research, 2025.

[6] “What’s Wrong With Low-Code Development Platforms? An Empirical Study of Low-Code Development Platform Bugs,” IEEE Transactions on Reliability, vol. 73, no. 1, pp. 695–709, 2024. doi: 10.1109/TR.2023.3295009.

[7] A. Bragança, I. Azevedo, N. Bettencourt, C. Morais, D. Teixeira, and D. Caetano, “Towards Supporting SPL Engineering in Low-Code Platforms Using a DSL Approach,” Proceedings of the 20th ACM SIGPLAN International Conference on Generative Programming: Concepts and Experiences, pp. 16–28, 2021. doi: 10.1145/3486609.3487196.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist 35

Unit	Zoho Creator Feature	Evidence	Where in the Application	Report Section	Figure / Code No.
I	Application created and navigation organised	Yes	All main modules visible	5.1	Fig. 5.1
II	Student and staff forms and list views	Yes	Student Details; Staff Details	5.2	Fig. 5.2–5.5
II	Hostel room form and list	Yes	Hostel Rooms; All Hostel Rooms	5.2–5.3	Fig. 5.6–5.7
III	Maintenance complaint form and list views	Yes	Maintenance Complaints; All Complaints	5.4–5.5	Fig. 5.8–5.9
IV	Feedback form and list	Yes	Complaint Feedback	5.5	Fig. 6.1; Fig. 8.4
V	Dashboard summary cards and charts	Yes	Hostel Admin Dashboard	5.4; 8.1	Fig. 5.1; Fig. 8.1
VI	Custom Deluge workflows	Proposed only	Not verified in application	5.5; Appendix C	DS-01 to DS-04
VII	External integrations / notifications	Not evidenced	Not shown	5.7	—
VIII	Mobile portal access / production deployment	Not evidenced	Development environment shown	7.1	—

Total feature groups evidenced in the supplied screenshots: 7. Proposed Deluge examples and unverified integrations are not counted as implemented features.

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters 36

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles and business value of low-code application development.	Ch. 1, 2	Good	
CO2	Develop data-driven applications using forms, reports and dashboards.	Ch. 4, 5, 6	Good	
CO3	Understand integrations and workflow automation in low-code platforms.	Ch. 5, 7	Satisfactory	
CO4	Consider deployment, security and governance requirements.	Ch. 7	Satisfactory	
CO5	Evaluate the scope and limitations of low-code application development.	Ch. 8, 9	Good	

Self-assessment scale: Excellent, Good, Satisfactory or Needs Improvement. The Examiner Level column is left blank for the examiner. Basis for the self-assessment:

- **CO1 – Good: the principles, architecture and justification for low-code development are covered in Chapters 1 and 2.**
- **CO2 – Good: the project demonstrates forms, list reports, dashboard components and complaint tracking views in Zoho Creator.**
- **CO3 – Satisfactory: the project identifies the role of integrations and workflow automation, but no external API, connection or notification workflow is evidenced in the supplied screenshots.**
- **CO4 – Satisfactory: the application was built and tested in a single development environment; deployment, audit and governance practices are described but not demonstrated.**
- **CO5 – Satisfactory: the limitations of low-code and the no-code to pro-code spectrum are analysed; no AI-assisted or pro-code component was used.**

APPENDIX C – DELUGE SCRIPT EXAMPLES

C.1 Hostel Room Capacity Validation

Location: Hostel Rooms form → On Validate workflow. Assumed field link names: Room_Capacity and Occupied_Count.

```
if(input.Room_Capacity < 0 || input.Occupied_Count < 0)
{
    alert "Capacity and occupied count cannot be negative.";
    cancel submit;
}

if(input.Occupied_Count > input.Room_Capacity)
{
    alert "Occupied count cannot exceed room capacity.";
    cancel submit;
}
```

Expected result: the room form is blocked when capacity or occupied count is negative, or when occupied count is greater than room capacity.

C.2 Set Complaint Date and Initial Status

Location: Maintenance Complaints form → On Validate workflow. Assumed field link names: Complaint_Date and Status.

```
if(input.Complaint_Date == null)
{
    input.Complaint_Date = zoho.currenttime;
}

if(input.Status == null || input.Status == "")
{
    input.Status = "Submitted";
}
```

Expected result: a new complaint receives the current date/time and Submitted status when the fields are empty.

C.3 Record the Resolved Date

Location: Maintenance Complaints form → On Validate workflow. Assumed field link names: Status and Resolved_Date.

```
if(input.Status == "Resolved")
{
    if(input.Resolved_Date == null)
    {
        input.Resolved_Date = zoho.currentdate;
    }
}
else
{
    input.Resolved_Date = null;
}
```

Expected result: the resolved date is populated when the complaint is marked Resolved. Clearing the date when status changes back is a suggested rule; retain it instead if audit history is required.

C.4 Complaint Feedback Rating Validation

Location: Complaint Feedback form → On Validate workflow. Assumed field link name: Rating. This example assumes Rating is a Number field.

```
if(input.Rating < 1 || input.Rating > 5)
{
    alert "Please enter a rating between 1 and 5.";
    cancel submit;
}
```

Expected result: feedback is accepted only when the rating is between 1 and 5.

Important: The screenshots do not expose the actual Zoho Creator field link names or confirm that these scripts have been configured. Replace assumed link names with the actual names and test each workflow before describing it as implemented.

These scripts are examples for form-level validation and date handling. Automatic staff allocation, email/SMS notifications and external integrations are not included in the demonstrated application.

APPENDIX D – USER MANUAL

D.1 Accessing the Application

The Hostel Maintenance Management System is a Zoho Creator application. It is accessed through a web browser using an account authorised by the application owner.

1. **Open Google Chrome, Microsoft Edge or Mozilla Firefox and sign in to the Zoho Creator account that has access to the application.**
2. **Open the Hostel Maintenance Management System from the Zoho Creator workspace.**
3. **Use the navigation menu to open Student Details, Staff Details, Hostel Rooms, Maintenance Complaints, Complaint Feedback or Hostel Admin Dashboard.**
4. **The supplied screenshots demonstrate the application in desktop view. Mobile access should be verified separately in the Zoho Creator application.**
5. **When finished, sign out of the Zoho account using the profile menu.**

D.2 Instructions for the Administrator

1. **To add a student, open Student Details and enter the register number, email, gender, department, year, hostel block, room number and phone number. Select Submit and verify the record in All Student Details.**
2. **To add staff, open Staff Details and enter the first name, last name, phone, email, staff category and availability. Submit the form and review the record in Staff Details List.**
3. **To maintain room information, open Hostel Rooms and enter the hostel block, floor number, room number, room capacity, occupied count, room status and assigned student details. Complete inspection fields where applicable.**
4. **Open Hostel Admin Dashboard to review Total Complaints, Pending Complaints, In Progress, Resolved Complaints, Total Students, Total Staff and the category-wise maintenance chart.**

5. **Open Maintenance Complaints to record a new issue. Enter the student, hostel block, room number, complaint category, description and priority. Add a photo if required and complete assignment and status fields where applicable.**
6. **Use All Maintenance Complaints, All Complaints, Pending Complaints and Resolved Complaints to review the complaint records and their status.**
7. **Restrict access to authorised users and review the permissions configured in Zoho Creator before using real student or staff data.**

D.3 Instructions for the User

1. **Open Maintenance Complaints from the navigation menu and select the add (+) control to open the complaint form.**
2. **Enter the student name, hostel block, room number, complaint category, complaint title and description. Add a complaint photo if needed.**
3. **Select the appropriate priority and complete the complaint date and other available details.**
4. **Select Submit. The complaint record should then be available in the complaint list views.**
5. **The supplied screenshots do not demonstrate automatic email or SMS notification after complaint submission.**
6. **Open All Complaints or All Maintenance Complaints to review the saved complaint records.**
7. **Use Pending Complaints, Resolved Complaints, Category Wise Complaints and Student Complaint History to review the records in different views.**
8. **After the maintenance work is completed, update the complaint status and resolution information using the available application fields, subject to the configured permissions.**